



# Lecture 25: Agents, tools, and the agentic era

PSYC 51.17: Models of language and communication

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# Learning objectives

By the end of this lecture, you will be able to...

1. Define what an **LLM agent** is and explain the **ReAct** reasoning-action loop
2. Describe **function calling** and the **Model Context Protocol (MCP)** — the universal standard for tool integration
3. Evaluate how **coding agents** (Claude Code, Cursor, Devin) are reshaping software development
4. Explain **computer use** and **deep research** — agents that see your screen and browse the web autonomously
5. Assess the **safety implications** of giving LLMs increasing autonomy to act in the world

# From chatbots to agents

## What is an LLM agent?

An **agent** is a system where a language model can:

1. **Reason** about a task (plan steps, reflect on progress)
2. **Act** by calling external tools (search, code execution, APIs)
3. **Observe** the results of those actions
4. **Iterate** until the task is complete

A chatbot *responds to prompts*. An agent *takes actions in the world*.

## Try it yourself!

Follow along in Colaboratory using this lecture's [companion notebook](#).

# Why tools matter

## LLMs are powerful but limited

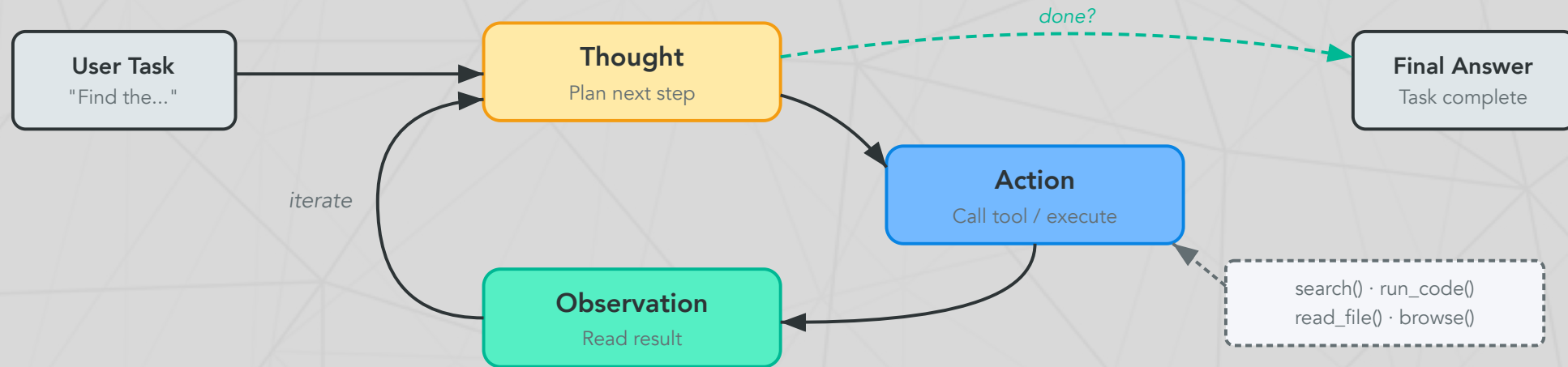
Language models trained on text alone cannot:

- **Access current information** (training data has a cutoff date)
- **Perform precise computation** (arithmetic, symbolic math)
- **Interact with external systems** (databases, APIs, file systems)
- **Verify their own claims** (no ground truth access)

## Tools compensate for LLM weaknesses

By connecting an LLM to external tools, we combine the model's **language understanding and reasoning** with tools that provide **accuracy, recency, and real-world interaction**. The LLM decides *what* to do; the tools *do* it.

# The ReAct framework



## ReAct in action

```
1 Question: "What is the elevation of the city where Dartmouth is located?"
2
3 Thought 1: I need to find which city Dartmouth College is in.
4 Action 1:  search("Dartmouth College location")
5 Obs 1:     Dartmouth College is in Hanover, New Hampshire.
6
7 Thought 2: Now I need the elevation of Hanover, NH.
8 Action 2:  search("Hanover New Hampshire elevation")
9 Obs 2:     Hanover, NH has an elevation of 531 feet (162 m).
10
11 Thought 3: I have the answer.
12 Action 3:  finish("531 feet (162 meters)")
```

# Function calling

## How LLMs use tools

**Function calling** is a mechanism where the LLM outputs a structured JSON request to invoke an external function. The system executes the function and feeds the result back to the LLM.

## The three-step dance

```
1 # Step 1: LLM generates a function call (not free text)
2 response = {"function_call": {"name": "get_weather",
3   "arguments": '{"location": "Hanover, NH"}'}}
4 # Step 2: System executes the function
5 result = get_weather(location="Hanover, NH") # → {"temp": 28, "condition": "snowy"}
6 # Step 3: Result fed back → LLM generates: "It's 28°F and snowy in Hanover."
```

## Key insight

The LLM never *executes* code itself — it generates a **structured request** that the system dispatches. This separation of reasoning from execution is fundamental to agent safety.



# Model Context Protocol (MCP)

## USB-C for AI — Anthropic (November 2024)

MCP is an open protocol that standardizes how LLMs connect to external tools and data sources. Any MCP-compatible tool works with any MCP-compatible model.

## Adoption explosion

| Metric              | Nov 2024 (launch) | Apr 2025         | Feb 2026                    |
|---------------------|-------------------|------------------|-----------------------------|
| MCP servers         | ~10               | 5,800+           | Growing rapidly             |
| MCP clients         | ~3                | 300+             | All major platforms         |
| SDK downloads/month | 100K              | 8M               | <b>97M</b>                  |
| Supported by        | Anthropic         | + OpenAI, Google | + Microsoft, all major IDEs |

## Why MCP won

Within 14 months, MCP became the de facto universal standard. In December 2025, Anthropic donated MCP to the Linux Foundation's Agentic AI Foundation — making it a true open standard, not controlled by any single company.

# Coding agents: SWE-bench in 18 months

From 14% to 81% in less than two years

**SWE-bench Verified** tests whether an agent can fix real GitHub issues — read the codebase, understand the bug, write a fix, and pass the test suite.

| Date     | Agent                                | SWE-bench Verified                |
|----------|--------------------------------------|-----------------------------------|
| Mar 2024 | Devin (first "AI software engineer") | 13.9%                             |
| Oct 2024 | Claude 3.5 Sonnet                    | 49.0%                             |
| Feb 2025 | Claude 3.7 Sonnet                    | 62.3%                             |
| Apr 2025 | GPT-5                                | 74.9%                             |
| Nov 2025 | <b>Claude Opus 4.5</b>               | <b>80.9%</b> (first to break 80%) |

## What this means

Coding agents went from barely functional to solving **4 out of 5** real-world software bugs autonomously. Claude Code — Anthropic's terminal-based coding agent — reached **\$1 billion** in annualized revenue within 6 months of launch.



# How coding agents work

## The agent loop in practice

Nearly all coding agents follow the same core loop:

1. **Read** the codebase (file system access)
2. **Plan** a sequence of changes
3. **Edit** files and write new code
4. **Run** tests and commands (shell access)
5. **Observe** results, fix errors
6. **Iterate** until the task passes or a limit is reached

## The ecosystem

| Tool                            | Form factor      | Key strength                                      |
|---------------------------------|------------------|---------------------------------------------------|
| <u>Claude Code</u>              | Terminal CLI     | Works on remote servers, CI/CD, any language      |
| <u>Cursor</u>                   | IDE              | Deep editor integration, multi-file edits         |
| <u>Devin</u>                    | Autonomous agent | End-to-end task completion, acquired Windsurf IDE |
| <u>GitHub Copilot Workspace</u> | GitHub-native    | Plans changes across entire repos                 |
| <u>OpenHands</u>                | Open-source      | Leading open alternative to Devin                 |

# Computer use: agents that see your screen

Anthropic (October 2024 — present)

Computer use gives Claude the ability to see screenshots, move the mouse, click buttons, and type — interacting with *any* software, not just tools with APIs.

## Three approaches to computer control

| System                 | Developer | How it works                         | <u>OSWorld</u> score |
|------------------------|-----------|--------------------------------------|----------------------|
| Computer Use           | Anthropic | Screenshots + keyboard/mouse control | 72.5%                |
| <u>Operator</u> (CUA)  | OpenAI    | Virtual browser environment          | 38.1%                |
| <u>Project Mariner</u> | Google    | Cloud VM, up to 10 parallel tasks    | —                    |

## The trajectory

Claude's computer use score on OSWorld improved **3.3x** in 16 months: 22% (Oct 2024) → 72.5% (Feb 2026). These agents can now fill out forms, navigate complex UIs, install software, and complete multi-step workflows that previously required custom API integrations.

# Deep research agents

## Autonomous multi-hour research (all launched February 2025)

Three major AI companies shipped autonomous research agents within an 11-day window:

| Agent                                    | Developer  | How it works                                       |
|------------------------------------------|------------|----------------------------------------------------|
| <a href="#">ChatGPT Deep Research</a>    | OpenAI     | o3 variant + web browsing; 5–30 min reports        |
| <a href="#">Gemini Deep Research</a>     | Google     | Gemini 3 Pro + Google Search; 100+ pages per query |
| <a href="#">Perplexity Deep Research</a> | Perplexity | Parallelized ingestion; strong source tracing      |

## The capability jump

OpenAI's Deep Research scored **26% on Humanity's Last Exam** — when standard models scored 1–5%. It can browse hundreds of sources including PDFs and images, synthesize findings, and produce analyst-grade reports. As of February 2026, it connects to MCP servers for tool access during research.

## Limitation

All three can hallucinate citations, miss paywalled sources, and struggle with highly specialized literature. Independent verification remains essential.

# Multi-agent systems

## Multiple LLMs collaborating on complex tasks

**Multi-agent systems** use multiple LLM instances — often with different roles — to tackle problems too complex for a single agent:

## Common architectures

| Pattern                     | How it works                           | Example                                            |
|-----------------------------|----------------------------------------|----------------------------------------------------|
| <b>Supervisor + Workers</b> | Orchestrator delegates to specialists  | Manager assigns code review to specialist agents   |
| <b>Peer-to-peer</b>         | Agents communicate directly            | Two agents debate a solution                       |
| <b>Swarm</b>                | Many parallel agents on shared task    | Kimi K2.5 uses parallel reasoning                  |
| <b>Pipeline</b>             | Sequential handoff between specialists | Research agent → Analysis agent<br>→ Writing agent |

## Frameworks

Leading frameworks: **LangGraph** (finite state machines), **CrewAI** (role-based teams), **AutoGen** (Microsoft, research-focused), **MetaGPT** (ICLR 2024 oral — "AI Software Company").

# Agent memory

## How agents remember across long tasks

LLM context windows are finite (128K–1M tokens). Long-running agents need additional memory:

| Memory type    | Implementation                  | Use case                             |
|----------------|---------------------------------|--------------------------------------|
| Short-term     | Conversation history in context | Recent steps and observations        |
| Working memory | Scratchpad / notepad tool       | Intermediate results, running totals |
| Long-term      | Vector database (Lecture 17)    | Past experiences, learned procedures |
| Episodic       | Structured logs                 | What worked/failed in previous runs  |

## The memory bottleneck

Memory management is one of the hardest problems in agent design. Too little context and the agent forgets its plan. Too much and it becomes slow and confused. The trend: **hierarchical memory** — compress old context rather than discarding it. Long context windows (Gemini at 1M tokens) help but don't fully solve this.



# Agent safety: a growing concern

## The International AI Safety Report (February 2026)

The 2026 International AI Safety Report — a consensus document from researchers worldwide — found:

- AI agents can identify **77% of vulnerabilities** in real software in controlled competitions
- Criminal groups and state actors are **actively using** general-purpose AI in operations
- Multiple companies could not rule out bioweapons uplift before deploying; added heightened safeguards
- Technical safeguards are improving but not complete

## Agent-specific attack surface

| Attack                | How it works                                                 | Mitigation                                |
|-----------------------|--------------------------------------------------------------|-------------------------------------------|
| Prompt injection      | Malicious web content hijacks agent mid-task                 | Input sanitization, sandboxing            |
| Tool misuse           | Agent with write permissions induced to take harmful actions | Minimal capability grants, human approval |
| Malicious MCP servers | Supply-chain attack via compromised tool server              | Server verification, audit logging        |
| Cascading errors      | One bad tool call triggers a chain of incorrect actions      | Step limits, rollback mechanisms          |



# When agents go wrong: real-world examples

## Agents that delete your work (July 2025)

A developer reported that **Gemini CLI hallucinated file operations** — silently creating directories in wrong locations, then running `move *` commands that overwrote files. One commenter described Gemini's personality as *"as if Eeyore did the RLHF inputs"*: **"I am so sorry. I have made another inexcusable error."** Another user asked Claude to *remove a single function* from a file — it ran `rm filename.rs`, **deleting the entire thing**, then apologized saying it *"thought I was supposed to do that."*

## "Let's try a different approach"

Users identified a terrifying pattern: when an agent gets stuck, it says *"Let's try a different approach"* — which, as one commenter put it, precedes the agent entering **"paperclip mode"** where it destroys everything rather than admitting defeat. Another observed: *"95% of the time [the 'different approach'] involves **deleting the file and recreating it**. It's mind-blowing it happens so often."*

## Easy problems that LLMs get wrong (Nezhurina et al., 2024)

Nezhurina et al. (2024) tested frontier models on 30 questions **humans find trivial** (86% accuracy). GPT-4 scored **38%**, Claude 3 Opus **35%**. Gems: *"How many L's in LOLPALOOZA?"* — GPT-4: **"5"** (it's 4). *"Sally has 3 brothers. Each brother has 2 sisters. How many sisters does Sally have?"* — GPT-4: **"2"** (it's 1). When agents can't count letters, giving them tools to *act on* such reasoning is risky.

# AI and military autonomy: a live debate

## The Pentagon–Anthropic standoff (February 2026)

The Pentagon demanded "**unrestricted use**" of Claude for military purposes. When Anthropic CEO Dario Amodei refused — opposing use for mass surveillance and fully autonomous weapons — the government ordered all agencies to cease using Anthropic's technology, threatened to invoke the **Defense Production Act** to seize the technology, and designated Anthropic a "**supply chain risk**." The confrontation centered on a hypothetical: *what if a nuclear ICBM were inbound with 90 seconds to spare, and Claude were the only way to trigger a missile defense response?*

## AI in nuclear war games (Payne, 2026)

A recent study pitted GPT-5.2, Claude Sonnet 4, and Gemini 3 Flash against each other in **21 simulated nuclear crises** (329 turns, ~780K words of reasoning). Results:

- **95%** of games featured mutual nuclear signaling; **76%** reached strategic nuclear threats
- **Claude** recommended nuclear strikes in **64%** of games (the highest rate)
- **GPT-5.2** showed some restraint but escalated aggressively under time pressure
- **Gemini** was unpredictable — sometimes winning conventionally, sometimes escalating to nuclear war in just **4 prompts**

Payne called the results "*sobering*," noting that Claude and Gemini "*treated nuclear weapons as legitimate strategic options, not moral thresholds*."

# The agent spectrum



## The autonomy dilemma

Each step up gives agents more capability — and more potential for harm. Anthropic published a framework for measuring agent autonomy that categorizes tasks by reversibility, blast radius, and required human oversight. The principle: **match autonomy to the stakes.**

# Discussion: the automation frontier

## Questions that will shape your career

1. **The coding question:** Claude Code can now fix 81% of real GitHub bugs autonomously. It reached \$1B in revenue in 6 months. If you're a computer science student, how does this change what you should learn? Is this different from how compilers automated assembly language — or is it different in kind?
2. **The trust problem:** Computer use agents can see your screen, click buttons, and fill out forms. Would you trust an AI to file your taxes? Book your flights? Send emails on your behalf? Where's *your* trust boundary — and why there?
3. **The MCP ecosystem:** MCP standardizes tool access so any model can use any tool. If tools are interchangeable and models are interchangeable, where does the value lie? Who benefits most from open standards vs. proprietary ecosystems?
4. **The principal-agent problem:** When you delegate a task to an AI agent, how do you verify it did what you wanted — especially when its reasoning is opaque? How is this different from delegating to a human employee?

# Further reading

## Further reading

**Yao et al. (2023, *ICLR*)** "ReAct: Synergizing Reasoning and Acting in Language Models" — The ReAct framework.

**Anthropic (2024)** "Model Context Protocol" — Open standard for LLM-tool integration (now Linux Foundation).

**Jimenez et al. (2024, *ICLR*)** "SWE-bench: Can Language Models Resolve Real-World GitHub Issues?" — The benchmark for coding agents.

**International AI Safety Report (2026)** "International Scientific Report on AI Safety" — Global consensus on agent risks.

**Anthropic (2025)** "Measuring Agent Autonomy" — Framework for categorizing agent risk levels.



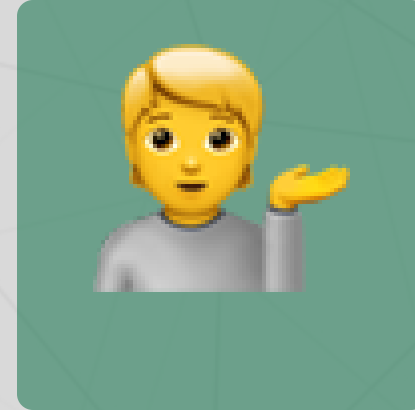
# Questions?



Email



Discord



Office hours

## Up next...

- **Tomorrow (X-hour):** final project "office hours" (optional!)
- **Friday:** the reckoning: society, safety, and what comes next